

QA Test Case: Verification of Germline Variant Classification for BRAF:V600E (hg38)

Objective

To confirm that the **Germline Variant Classification** for **BRAF:V600E** (reference genome **hg38**) is correctly displayed as “**Pathogenic**”, with the expected score and interpretation, after entering specific sample information in the **Optional Sample Information** modal and navigating to the detailed results page.

Detailed Test Steps

Step 1 – Launch VarSome Website

1. Open a web browser and navigate to <https://varsome.com>.
2. Wait for the homepage to fully load.

Expected Result:

The VarSome homepage is displayed, featuring a central search bar labeled “**Enter gene, transcript, variant, or region.**”

Step 2 – Initiate Variant Search

1. In the search bar, type **BRAF:V600E**.
2. In the dropdown next to the search bar, verify that the reference genome is set to **hg38**.
3. Click the **Search** button.

Expected Result:

A modal titled “**Optional Sample Information**” appears automatically before the variant results page is loaded.

Step 3 – Complete the Optional Sample Information Modal

1. Ensure the **Germline** tab is selected (the tab header should appear highlighted in green).
2. Fill in the fields as follows:
 - **Phenotype:** Type “Cancer” and select **Cancer (MONDO:0004992)** from the dropdown list.
 - **Sex:** Select **Female**.
 - **Age at onset:** Enter **60**.
 - **Ethnicity:** Select **East Asian** from the dropdown list.
3. Leave the following fields blank: **Inheritance**, **Family Members Affected**, **Zygosity**, and **Family Segregation**.
4. Click the **Search** button at the bottom of the modal.

Expected Result:

The modal closes, and the system redirects to the detailed results page for **BRAF:V600E (hg38)**, applying the specified Germline sample information.

Step 4 – Verify the Results Page Load

1. Wait for the results page to fully load.
2. Confirm that the page displays detailed variant information, including:
 - **General Information**
 - **Germline Classification**
 - **PharmKGB**, **ClinVar**, **LOVD**, and **Publications** sections

Expected Result:

The results page is successfully populated with variant-specific data. A “**Germline Classification**” card is visible within the top information panel.

Step 5 – Expand the Germline Classification Section

1. Locate the “**Germline Classification**” card.
2. Click the card or the expand icon (arrow) to open the detailed view.
3. Scroll down to view the **Germline Variant Classification** section.

Expected Result:

The section expands to show automated evidence rules and the final classification verdict.

Step 6 – Verify the Classification Verdict

1. Observe the verdict displayed in the **Germline Variant Classification** header.

Expected Result:

The classification verdict is displayed as “**Pathogenic**” in red text.

The implementation should be carried out using **Python** in combination with **Selenium**.